Lab Assignment No.	03
Title	Implement Union, Intersection, Complement and Difference operations on fuzzy sets. Also create fuzzy relations by Cartesian product of any two fuzzy sets and perform max-min composition on any two fuzzy relations.
Roll No.	
Class	BE AI & DS
Date Of Completion	
Subject	Computer Laboratory III[417534]
Assessment Marks	
Assessor’s Sign	

Assignment No. 03

Title: Implementation of Fuzzy Set Operations and Fuzzy Relations

Problem Statement: Implement Union, Intersection, Complement, and Difference operations on fuzzy sets. Additionally, create fuzzy relations using the Cartesian product of any two fuzzy sets and perform max-min composition on two fuzzy relations.

Prerequisites: Basic understanding of fuzzy set theory, fuzzy relations, mathematical operations on fuzzy sets, max-min composition, and Python programming.

Software Requirements: Windows/Linux/MacOS, Python (NumPy, Pandas, Matplotlib), and SciPy library for mathematical computations.

Hardware Requirements: Intel Core i3 or higher, 4GB RAM, 200MB free disk space, and an active Python environment.

Theory:

Introduction to Fuzzy Sets:

A fuzzy set is an extension of a classical set where each element has a degree of membership between 0 and 1. Fuzzy sets are used in approximate reasoning and decision-making applications.

Fuzzy Set Operations:

1. **Union (A ∪ B):** The maximum membership value for each element from both sets.
   - Formula: $\mu(A \cup B) = \max(\mu A, \mu B)$
2. **Intersection (A ∩ B):** The minimum membership value for each element from both sets.
   - Formula: $\mu(A \cap B) = \min(\mu A, \mu B)$
3. **Complement (A'):** The degree of non-membership of each element.
   - Formula: $\mu(A') = 1 - \mu(A)$
4. **Difference (A - B):** Elements that belong to A but not to B.
   - Formula: $\mu(A - B) = \min(\mu A, 1 - \mu B)$

Fuzzy Relations and Cartesian Product:

A fuzzy relation is a mapping between two fuzzy sets using the Cartesian product.

- If A and B are two fuzzy sets, their Cartesian product results in a fuzzy relation R.
- The membership function of the relation is defined as:
  - $\mu R(A, B) = \min(\mu A(x), \mu B(y))$

Max-Min Composition of Fuzzy Relations:

- The max-min composition is used to derive a new fuzzy relation from two existing fuzzy relations.
- Formula:
  - $\mu R \circ S(x, z) = \max\_y (\min(\mu R(x, y), \mu S(y, z)))$

Implementation Approach:

1. **Define fuzzy sets:** Represent them as Python dictionaries or arrays.
2. **Perform operations:** Compute union, intersection, complement, and difference using NumPy functions.
3. **Create fuzzy relations:** Use the Cartesian product to form a relation matrix.
4. **Max-Min Composition:** Implement a function to compute max-min composition between two fuzzy relations.
5. **Visualization:** Use Matplotlib to plot fuzzy sets and relations.

Applications of Fuzzy Sets and Relations:

- Fuzzy logic in artificial intelligence and expert systems.
- Decision-making in uncertain environments.
- Control systems such as washing machines and air conditioning.
- Image processing and pattern recognition.

Source Code –

```
import numpy as np

# Function to perform Union operation on fuzzy sets

def fuzzy_union(A, B): return np.maximum(A, B)

# Function to perform Intersection operation on fuzzy sets

def fuzzy_intersection(A, B): return np.minimum(A, B)

# Function to perform Complement operation on a fuzzy set

def fuzzy_complement(A): return 1 - A

# Function to perform Difference operation on fuzzy sets

def fuzzy_difference(A, B): return np.maximum(A, 1 - B)

# Function to create fuzzy relation by Cartesian product of two fuzzy sets

def cartesian_product(A, B): return np.outer(A, B)

# Function to perform Max-Min composition on two fuzzy relations

def max_min_composition(R, S): return np.max(np.minimum.outer(R, S), axis=1)

# Example usage

A = np.array([0.2, 0.4, 0.6, 0.8]) # Fuzzy set A

 B = np.array([0.3, 0.5, 0.7, 0.9])# Fuzzy set B

# Operations on fuzzy sets

union_result = fuzzy_union(A, B)

intersection_result = fuzzy_intersection(A, B)

complement_A = fuzzy_complement(A)

difference_result = fuzzy_difference(A, B)

print("Union:", union_result)

print("Intersection:", intersection_result)

print("Complement of A:", complement_A)
```

```
print("Difference:", difference_result)

# Fuzzy relations

R = np.array([0.2, 0.5, 0.4]) # Fuzzy relation R

S = np.array([0.6, 0.3, 0.7]) # Fuzzy relation S

 # Cartesian product of fuzzy relations

cartesian_result = cartesian_product(R, S)

# Max-Min composition of fuzzy relations

composition_result = max_min_composition(R, S)

print("Cartesian product of R and S:") print(cartesian_result)

 print("Max-Min composition of R and S:")

print(composition_result)
```

Conclusion: This practical demonstrates the implementation of fuzzy set operations and fuzzy relations using Python. It highlights the importance of fuzzy logic in decision-making and real-world applications.